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B-Mamba: Burn severity mapping from large-scale Landsat imagery using a Mamba-based Siamese network

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***Abstract*— Operational burn severity mapping remains largely dependent on spectral index-based approaches that require manual threshold selection and expert interpretation, limiting scalability and consistency across events. Deep learning methods have improved change detection accuracy, but most have been developed for very high-resolution datasets, reducing their transferability to medium-resolution, globally available sensors such as Landsat. To address this gap of automating the operational burn severity mapping with deep learning using Landsat images, we propose burn severity Mamba (B-Mamba), a Siamese Mamba-based network that exploits the linear-complexity, long-range dependency modeling of state-space architectures while explicitly capturing pre- to post-fire changes through dual-branch feature extraction, hierarchical change fusion, and a progressive reconstruction decoder. On the large-scale Landsat-BSA benchmark dataset, B-Mamba achieves state-of-the-art performance, reaching a mean Intersection-over-Union (mIoU) of 79.31% and an overall accuracy of 95.70%, outperforming leading convolutional, transformer-based, and existing Mamba-based change detection models by 3.71–11.66% in mIoU. Ablation experiments confirm the importance of the learnable patch-expanding decoder, change fusion and the proposed composite loss function for accurate severity delineation. Additional evaluations on independent wildfire events within and outside the contiguous United States (Alaska and Hawaii), including evaluations on Sentinel-2 imagery, demonstrate the robustness and generalization of the approach. These results indicate that Mamba-based state-space architectures provide a scalable, effective pathway toward operational, large-area burn severity mapping from medium-resolution satellite imagery.**

***Index Terms*—Burned severity, VMamba, Landsat-BSA, Siamese network.**

I. INTRODUCTION

Wildfires are escalating not only in frequency and burned area, but also in the complexity of their impacts across landscapes and infrastructure systems [1]. In this context, burn severity mapping has shifted from a retrospective ecological product to an operational layer that informs risk management and recovery planning. Severity maps are used to anticipate post-fire erosion and debris-flow hazards, diagnose hydrologic regime shifts that threaten downstream assets, and support the prioritization of mitigation and restoration activities [2]. They also underpin estimates of wildfire-driven carbon emissions and associated updates to greenhouse gas inventories, which influence mitigation accounting from regional to national scales [3], [4]. Yet, despite the central role of burn severity products, the community still lacks an approach that is simultaneously accurate, scalable, and consistent across events.

Field-based methods such as the Composite Burn Index (CBI) remain reliable at plot scales, but they cannot provide the spatial coverage and throughput required under the current fire regime [5]. Consequently, multispectral satellite imagery has become the primary basis for large-area burn severity assessment [6], [7]. The pronounced spectral contrast between burned and unburned surfaces in the shortwave infrared (SWIR) and near-infrared (NIR) has led to widespread use of indices such as the Normalized Burn Ratio (NBR) and its derivatives, including differenced NBR (dNBR) and relativized dNBR (rdNBR) [8], [9]. These index-based approaches have enabled event-scale mapping across many regions [10]– [15] and have been operationalized by programs such as MTBS, which apply dNBR/rdNBR thresholds supported by expert interpretation and ground calibration [16]. However, this operational success comes with a persistent limitation: medium- and high-severity delineations can vary substantially with analyst decisions and event-specific conditions [17]. As a result, prevailing workflows are labor-intensive and difficult to standardize, which reduces reproducibility and complicates cross-event comparisons.

Deep learning has been proposed as a pathway toward more

This research was supported by the R&D Program for Forest Science Technology (RS-2024-00404017) funded by the Korea Forest Service (Korea Forestry Promotion Institute) and by Korea Institute of Marine Science & Technology (KIMST) funded by the Ministry of Oceans and Fisheries (RS-2023-00256330).

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consistent severity mapping and is now widely used across wildfire monitoring tasks, including susceptibility mapping, active fire detection, burned-area delineation, and post-fire recovery modeling [18]– [23]. For burn severity mapping specifically, U-Net variants can learn spatial context and produce structured segmentation beyond what fixed thresholds can achieve [24]– [26]. Yet, most prior work has focused on post-fire Sentinel-2 settings and architectural refinements within convolutional networks, including attention modules and residual connections [26], [27]. More recent studies have leveraged bi-temporal information through Siamese designs and auxiliary dNBR features [27], and large-scale evaluations suggest that attention-based U-Net models combined with Online Hard Example Mining can improve Landsat-based severity classification [28]. Despite these advances, two challenges remain: (i) scalability for continental-scale mapping with long image sequences and large spatial footprints, and (ii) robustness to fragmented, spectrally mixed burn patterns that are intrinsic to medium-resolution imagery.

Transformer-based architectures address part of this problem by strengthening long-range context modeling. For example, [29] reported that a modified MapFormer model [30], augmented with auxiliary variables such as NDVI, canopy cover, and slope, can better capture complex spectral–spatial patterns than CNNs. However, these gains often come with practical constraints. Combining all Landsat spectral bands with multiple auxiliary inputs increases compute and memory demands, and self-attention scales quadratically with input size [31], creating bottlenecks for large-area operational mapping. Thus, improvements achieved by high-capacity attention models may not translate into methods that can be deployed at scale with stable resource requirements.

This gap motivates alternative architectures that enable long-range modeling without quadratic scaling. Structured state-space sequence (S4) models provide linear-time sequence modeling with competitive representational capacity [32]. VMamba extends this paradigm to visual representations and has demonstrated strong performance across core vision tasks [33]. Recent Mamba-based models, including Change Mamba [31] and DiffMamba [34], further suggest that state-space approaches can be effective for remote sensing change detection and segmentation. However, much of the evidence to date has come from very high-resolution, urban-centric benchmarks (e.g., LEVIR-CD, OpenEarthMap, and SECOND), where changes often exhibit crisp geometry and limited spectral mixing. Burn severity mapping differs fundamentally from these. It relies on medium-resolution sensors (Landsat/Sentinel-2), where sub-pixel mixing is common, and burn severity patterns are irregular, fragmented, and strongly modulated by vegetation, topography, and phenology [35], [36]. These characteristics define the operational regime rather than constituting edge cases. Medium-resolution imagery remains indispensable because it offers global coverage, frequent revisit, and open access required for sustained monitoring [37]. Accordingly, it is important to evaluate whether modern state-space Mamba architectures can deliver both accuracy and scalability under the constraints faced in operational burn severity mapping.

In this study, we address this need by systematically investigating Mamba-based state-space models for large-scale burn severity mapping using medium-resolution Landsat imagery. We propose a Siamese Mamba-based network, termed **B-Mamba**, designed to explicitly model bi-temporal changes between pre- and post-fire observations and to better represent fragmented burn scars across heterogeneous landscapes. We further evaluate Mamba's capacity to capture long-range spatial dependencies and complex spectral-spatial change patterns through comprehensive comparisons against convolutional and state-of-the-art deep learning baselines. Together, these experiments clarify the practical trade space between accuracy and scalability and support the development of robust, deployable methods for operational wildfire impact assessment and post-fire monitoring.

II. Methodology

A. Study area and dataset

This study uses the Landsat-BSA dataset [28] as the primary benchmark for evaluating deep learning-based burn severity mapping. Landsat-BSA consists of paired pre-fire and post-fire top-of-atmosphere (TOA) images acquired from the Landsat series (Landsat 5, Landsat 7, and Landsat 8), together with reference burn severity maps derived from the Monitoring Trends in Burn Severity (MTBS) project. Each sample covers 256×256 pixels and includes three spectral bands of shortwave infrared 2 (SWIR2), near-infrared (NIR), and shortwave infrared 1 (SWIR1), which are widely used for burn severity assessment due to their sensitivity to post-fire spectral changes.

The 256×256 patch size was adopted for three reasons: (i) the source dataset [28] is natively distributed at this resolution; (ii) this size is widely used to benchmark state-of-the-art segmentation and change detection models in remote sensing, enabling fair and consistent comparison with existing baselines; and (iii) larger patches substantially increase computational demand, since the selective-scan operations in the Mamba architecture scale with sequence length, making larger inputs impractical under the available memory budget. In preliminary experiments, we also evaluated 128×128 patches, which produced results broadly comparable to those obtained at 256×256. Given this similar performance, we retained 256×256, as it is the native resolution of the Landsat-BSA benchmark and the patch size most widely used in related burned-area and fire remote-sensing work, supporting consistency and comparability with prior studies. Constructing 512×512 samples would require merging adjacent 256×256 patches, which may combine neighboring patches with different spatial characteristics and alter the original sample structure, while also substantially increasing memory and training costs. Therefore, 256×256 provides a practical balance between comparability with existing work, computational efficiency, and fidelity to the original Landsat-BSA benchmark, and was retained for all experiments.

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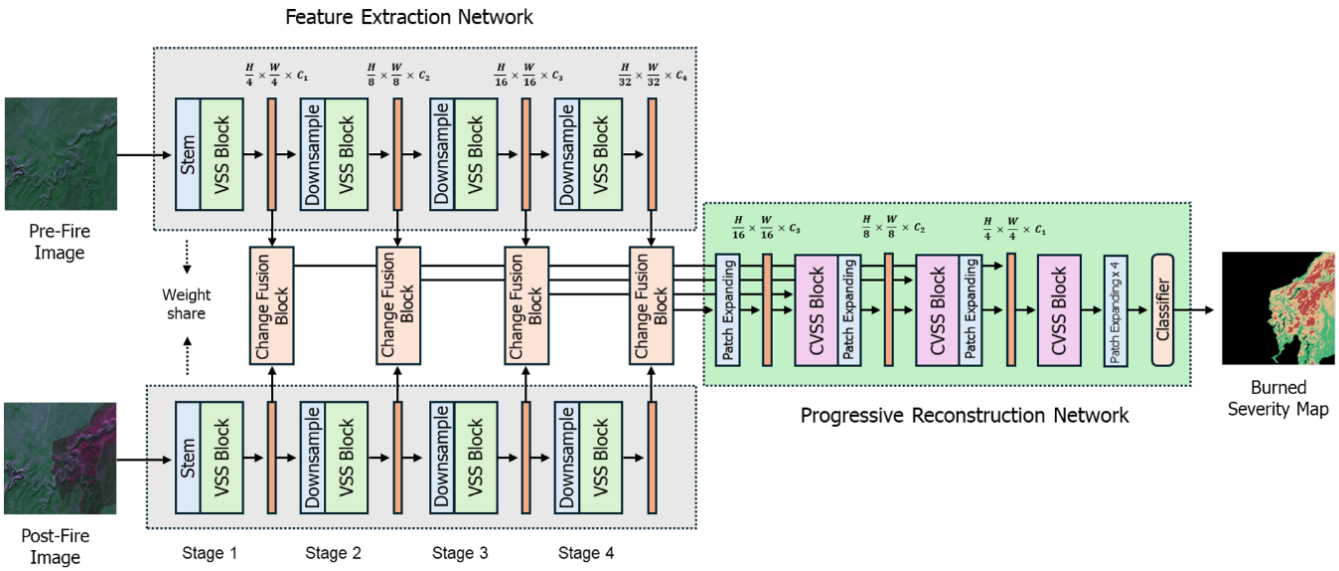


Fig 1. Model architecture of B-Mamba. The model integrates a dual-branch architecture that fuses pre- and post-fire images and comprises three main components: (1) feature extraction network to encode pre- and post-fire imagery; (2) change fusion blocks to efficiently merge features from both encoder branches; and (3) progressive reconstruction network to decode and reconstruct burn severity masks from the fused features.

Covering wildfire events from 2010 to 2019, Landsat-BSA spans nearly all states in the conterminous United States (Fig. 3). The dataset contains 6,000 samples representing diverse fire regimes, vegetation types, and seasonal conditions, enabling a comprehensive evaluation of burn severity mapping methods. We adopt the original dataset split of 70%/20%/10% for training/validation/testing. This stratified partition aims to balance representation across regions and events, reducing sampling bias and supporting robust generalization. Throughout this study, unburned, low severity, moderate severity, high severity, and non-processing area/cloud are encoded as 0, 1, 2, 3, and 4, respectively, in the ground-truth labels. The five classes are strongly imbalanced, with unburned pixels dominating the dataset. The class proportions for the training, validation, and test sets are adopted from [28] and presented in Table I. These values support the class-weighting and composite loss function described in Section II.F.

TABLE I

PER-CLASS PIXEL DISTRIBUTION (PERCENTAGE OF TOTAL LABELED PIXELS) ACROSS THE TRAINING, VALIDATION, AND TEST SPLITS OF THE LANDSAT-BSA DATASET.

Class	Train (%)	Val (%)	Test (%)
Unburned (0)	72.55	74.31	74.46
Low severity (1)	16.28	15.15	15.17
Moderate severity (2)	6.81	6.36	6.44
High severity (3)	3.15	3.13	2.91
Non-processing/cloud (4)	1.20	1.05	1.01

B. Model overview

The proposed burn-severity Mamba model (B-Mamba) is built upon VMamba and adopts a Siamese encoder-decoder framework inspired by Sigma [38] (Fig. 1). Rather than introducing a new backbone architecture, B-Mamba adapts these existing components for large-scale burn severity mapping through task-specific modifications. The architecture comprises (i) a dual-branch encoder for feature extraction from pre- and post-fire imagery, (ii) change fusion blocks that integrate bi-temporal information at multiple feature scales, and (iii) a progressive decoder that reconstructs burn severity masks from fused representations.

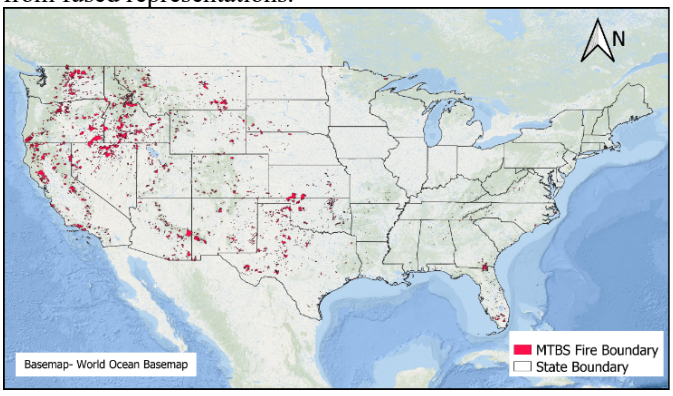


Fig 2. Spatial distribution of wildfire perimeters recorded by the MTBS dataset during 2010–2019 across the conterminous United States. These events were used to compile the Landsat-BSA dataset analyzed in this study.

Unlike the original VMamba and Sigma architectures, B-Mamba is designed to better capture fragmented and spatially heterogeneous burn-severity patterns. To this end, the decoder employs a patch-expanding strategy inspired by Swin Transformer–style upsampling rather than conventional bilinear interpolation, helping preserve fine-scale spatial structures that are important for delineating burn severity in heterogeneous landscapes.

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B-Mamba's contribution lies in three components specifically designed for the fragmented and spatially heterogeneous nature of burn severity patterns: (i) change fusion blocks that integrate bi-temporal information at multiple encoder scales through a cross-selective-scan mechanism, rather than Sigma's fusion strategy, to better capture scattered burn signatures; (ii) a Swin Transformer–style patch-expanding decoder that replaces bilinear interpolation and better preserves fine-scale spatial boundaries between burn severity classes; and (iii) a composite OHEM-WCE and Lovász–Softmax loss function that addresses the pronounced class imbalance characteristic of burn severity data. The ablation results in Table IV isolate the contribution of each design choice.

B-Mamba retains the computational efficiency of state-space models while strengthening long-range spatial modeling, which is important for large-scale burn severity mapping from medium-resolution satellite imagery. We implement three variants—tiny, small, and base—which differ in the number of encoder layers and feature-channel dimensions. Architectural details are summarized in Table II. The implementation of B-Mamba is publicly available at: https://github.com/sohel-raana/B-Mamba.

TABLE II
SUMMARY OF MODEL VARIANTS BY NUMBER OF LAYERS (L_X) AND FEATURE CHANNELS (C_X).

Stage	Tiny (B-Mamba-T)	Small (B-Mamba-S)	Base (B-Mamba-B)
1	$L_1 = 2$ $C_1 = 96$	$L_1 = 2$ $C_1 = 96$	$L_1 = 2$ $C_1 = 128$
2	$L_2 = 2$ $C_2 = 192$	$L_2 = 2$ $C_2 = 192$	$L_2 = 2$ $C_2 = 256$
3	$L_3 = 9$ $C_3 = 384$	$L_3 = 27$ $C_3 = 384$	$L_3 = 27$ $C_3 = 512$
4	$L_4 = 2$ $C_4 = 768$	$L_4 = 2$ $C_4 = 768$	$L_4 = 2$ $C_4 = 1024$

C. Feature extraction network (Encoder)

The encoder follows a Siamese configuration with two parallel branches sharing weights, ensuring consistent feature extraction from pre-fire and post-fire imagery. Each branch receives two input images I_{pre} and I_{post}, where $I \in \mathbb{R}^{H\times W\times 3}$. Feature extraction begins with a stem module inspired by Vision Transformer architectures, which partitions the input into patches and generates initial feature representations. This stem consists of a series of 2D convolution layers that extract low-level features, producing feature maps $F_i^1 \in \mathbb{R}^{H/4\times W/4\times C_1}$, where i ∈ {pre, post} denotes the respective input time.

The core component of the encoder is the Visual State Space Block (VSSB), which follows the Visual State Space paradigm and incorporates SS2D modules to model long-range spatial dependencies. Each VSSB includes a linear projection, a depth-wise convolution, and a residual SS2D module. The SS2D module decomposes feature maps into four directional sequences traversing complementary diagonal pathways: top-left to bottom-right, bottom-right to top-left, top-right to bottom-left, and bottom-left to top-right (Fig. 4). A dedicated selective scan state model is applied to each direction, learning direction-specific parameters (A, B, C, D, Δ). This multi-directional scanning strategy enables comprehensive modeling of spatial dependencies across the entire feature map, following the VMamba design [33].

The encoder comprises four VSSBs arranged hierarchically. Each block beyond the first is preceded by a downsampling module, forming a multi-scale feature extraction pipeline. This produces hierarchical feature representations $F_i^2 \in \mathbb{R}^{H/8\times W/8\times C_2}$, $F_i^3 \in \mathbb{R}^{H/16\times W/16\times C_3}$, and $F_i^4 \in \mathbb{R}^{H/32\times W/32\times C_4}$. Each VSSB concludes with layer normalization and a linear transformation to stabilize training and ensure proper feature conditioning. This hierarchical design combines the strengths of convolutional layers for local feature extraction with state-space models for long-range spatial modeling, making it well suited for bi-temporal burn severity analysis.

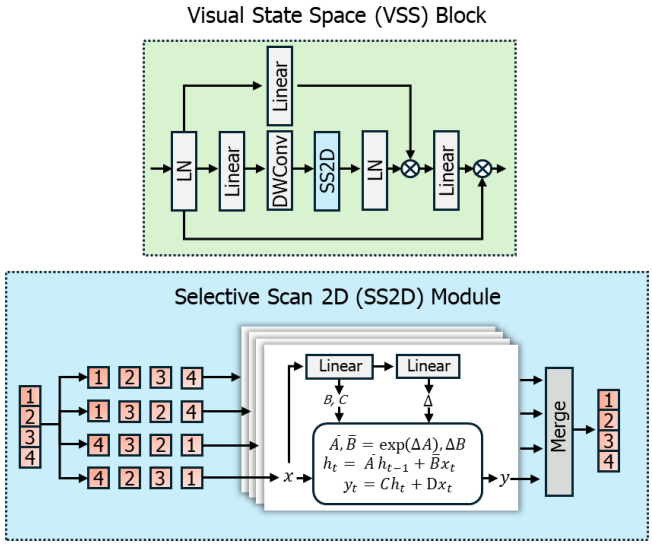


Fig 3. Structure of each VSS block of the encoder. Within a VSSB, the input image undergoes two-dimensional selective scanning, which encodes the image and emphasizes salient features.

D. Change fusion block

To integrate information from the dual-branch encoder, we introduce a change fusion block (CFB), illustrated in Fig 5. At the k-th encoder stage, the CFB operates on feature maps F_{pre}^k and F_{post}^k, each with dimensions $H_k \times W_k \times C_k$. The two feature maps are first independently projected using linear layers and refined via depth-wise convolution to preserve both local and global spatial information. The processed features are then passed to a cross selective scan (cross-scan) module, which explicitly models contextual dependencies between pre-fire and post-fire representations through cross-multiplication mechanisms.

Within the cross-scan module, learnable system matrices (B, C, Δ) are generated for each input. At each spatial location, the hidden state of the pre-fire feature is updated based on both its own input and the corresponding post-fire feature, and vice versa. This bi-directional selective scan process is conceptually analogous to cross-attention, but operates within a state-space

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framework, allowing efficient temporal interaction modeling. The outputs of the cross-scan module are two enriched feature maps, $\hat{F}_{pre}$ and $\hat{F}_{post}$, each containing information from both temporal states. To explicitly emphasize fire-induced changes, a subtraction operation is applied:

$$F_{diff} = \hat{F}_{pre} - \hat{F}_{post} \qquad (1)$$

This difference representation highlights burn-related spectral-spatial changes and serves as the primary input to the decoder.

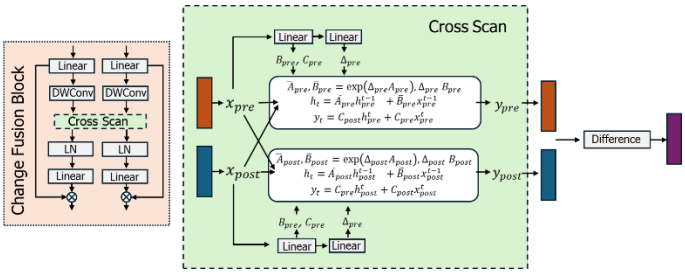


Fig 4. Structure and the cross-scan in the change fusion module. The fusion module incorporates both pre- and post-fire images and performs cross-scanning between them, effectively emphasizing their differences through the final output of the cross-scan operation.

E. Progressive reconstruction network (Decoder)

The progressive reconstruction network decodes the fused features into high-resolution burn severity maps. As shown in Fig 6, the decoder follows a U-Net–inspired design but replaces conventional upsampling with a patch-expanding mechanism. This approach restores spatial resolution using learnable parameters, enabling more accurate reconstruction of fine-scale spatial details.

Each decoding stage consists of a channel-aware visual state space block (CVSSB). The CVSSB first applies a VSSB to capture global spatial dependencies. To address inter-channel interactions that are not explicitly modeled by VSSBs, the CVSSB incorporates channel attention using average pooling and max pooling along the channel dimension. This design allows the decoder to emphasize salient spectral features relevant to burn severity discrimination.

Patch expanding progressively increases spatial resolution while maintaining compatibility with skip connections from the encoder. Two residual connections within each CVSSB facilitate gradient propagation and stabilize training. At each decoding level, expanded features are fused with the corresponding encoder features, enabling hierarchical context aggregation. After the final patch-expanding operation, a multi-layer perceptron (MLP) classification head predicts pixel-wise burn severity labels, producing the final segmentation map.

F. Loss functions

Burn severity datasets are typically characterized by strong class imbalance, with unburned pixels dominating the sample distribution. To address this issue, we adopt a composite loss function combining online hard example mining with weighted cross-entropy (OHEM-WCE) and the Lovász-Softmax loss.

The OHEM-WCE loss prioritizes hard-to-classify pixels while down-weighting easy samples [39]. For each mini-batch, pixel-wise cross-entropy is computed for all valid pixels, excluding ignored labels. Predicted probabilities for the ground-truth class are sorted in ascending order, and pixels with lower confidence are selected as hard examples. A minimum number of pixels is enforced, and a probability threshold of 0.5 ensures retention of uncertain samples. The OHEM-WCE loss is defined as:

$$\mathcal{L}_{OHEM-WCE} = \frac{1}{|H|} \sum_{i \in \mathcal{H}} W_{y_i} CE\,(p_i, y_i) \qquad (2)$$

where H is the set of selected hard pixels, p_i is the predicted logit, and y_i is the ground truth label.

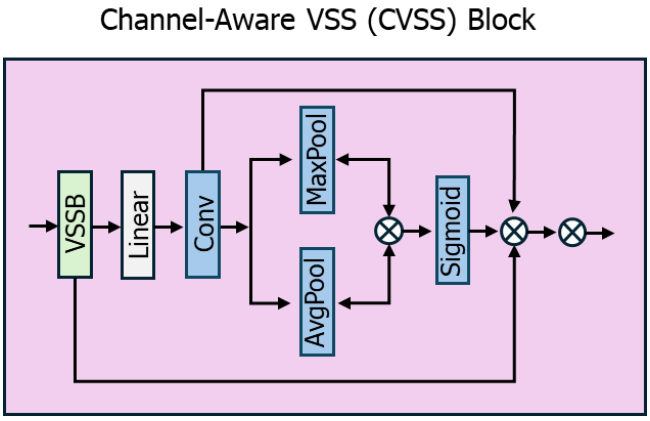


Fig 5. Structure of the channel-aware VSS (CVSS) block in the progressive reconstruction decoder network. The CVSS block adaptively models inter-channel dependencies through channel attention, while the state space mechanism captures long-range spatial dependencies. Together, these components enhance feature reconstruction and facilitate more accurate recovery of fine-scale details during decoding.

The class weights were determined specifically, for a dataset with N total pixels and n_c pixels belonging to class C, the weight for class C was defined as:

$$w_c = \frac{N}{C.n_c} \qquad (3)$$

To further improve boundary delineation, we incorporate the Lovász-Softmax loss, which directly optimizes the mean Intersection-over-Union (mIoU). The final loss function is defined as:

$$\mathcal{L}_{severity} = 0.3 \times \mathcal{L}_{OHEM-WCE} + 0.7 \times \mathcal{L}_{Lova'sz-Softmax} \qquad (4)$$

This loss function effectively addresses class imbalance, encouraging the model to learn subtle severity nuances and robust spatial representations.

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TABLE III

COMPARISON OF DIFFERENT MODELS FOR BURNED SEVERITY MAPPING. ALL VALUES ARE IN PERCENTAGE. * MARKED VALUE HAS TAKEN FOR BI-TEMPORAL INPUTS ONLY (NO ANCILLARY SPECTRAL INDICES) AS REPORTED IN [28]. THE BOLD VALUES SHOW THE BEST SCORE WHILE THE UNDERSCORE SHOWS THE SECOND-BEST RESULT

Model	mIOU	Kappa	OA	FWIoU	IoU per Class [0, 1, 2, 3, 4]	Infer. Time
Attn. U-Net*	75.41	87.66	94.82	91.41	[96.68, 77.73, 66.04, 72.66, 63.94]	-
Bi-AttnUNet	71.61	83.89	93.02	88.14	[94.99, 70.73, 62.51, 69.86, 59.96]	20.9
SNUNet	73.05	86.07	94.11	89.76	[96.32, 74.99, 62.19, 69.80, 61.95]	16.9
TransUNet	69.99	81.44	91.86	86.57	[94.30, 65.73, 57.93, 69.54, 62.45]	16.1
BIT	64.83	79.10	91.00	85.49	[94.72, 63.71, 48.28, 56.44, 61.10]	**1.71**
ChangeFormerV6	70.80	82.25	93.34	88.60	[95.85, 71.96, 57.93, 66.82, 61.44]	33.3
FTN	73.55	86.13	94.14	89.80	[96.36, 74.93, 61.94, 70.12, 64.40]	40.2
ConvFormer-CD	75.11	87.44	94.71	90.67	[96.67 77.29, 65.31, 72.06, 64.24]	4.38
Change Mamba-B	69.00	82.57	92.61	87.65	[95.76, 69.23, 52.71, 61.39, 65.91]	54.3
BiFA	69.55	83.60	93.09	88.27	[95.99, 70.96, 54.79, 64.28, 61.73]	3.59
CDMamba	75.60	87.66	94.80	90.82	[96.77, 77.72, 65.05, 72.38, 66.10]	15.8
HAM-CD	66.41	80.12	91.41	86.10	[94.89, 65.01, 49.98, 58.75, 63.38]	7.35
B-Mamba-T	78.73	89.44	95.56	91.98	[97.01, 80.53, 71.26, 77.24, 67.59]	50.8
B-Mamba-S	**79.31**	**89.77**	**95.70**	**92.21**	[**97.10**, 80.94, **72.33**, **78.09**, **68.06**]	70.6
B-Mamba-B	79.23	89.75	95.69	92.20	[**97.10**, **80.99**, 72.23, 77.82, 67.97]	92.2

G. Evaluation metrics

Model performance is evaluated using standard metrics for multi-class segmentation. Overall accuracy (OA) is used to assess pixel-wise classification accuracy across all classes. Mean Intersection-over-Union (mIoU) is employed as the primary metric for model comparison. Cohen's Kappa coefficient is calculated to quantify agreement between predictions and reference labels beyond chance. In addition, class-wise IoU and frequency-weighted IoU (FWIoU) are reported to account for class frequency effects. The corresponding formulations are given in Eqs. (5)– (9). Model selection is based primarily on mIoU.

$$\text{OA} = \sum_i^C \frac{\text{TP}_i}{(\text{TP}_i + \text{FP}_i + \text{FN}_i)} \quad (5)$$

$$\text{IoU} = \sum_i^C \frac{\text{TP}_i}{(\text{TP}_i + \text{FP}_i + \text{FN}_i)} \quad (6)$$

$$\text{mIoU} = \frac{1}{C} \times IoU \quad (7)$$

$$\text{FWIoU} = \frac{1}{\sum_{i=1}^C N_i} \times IoU \quad (8)$$

$$\text{Kappa} = \frac{2\times(\text{TP}\times\text{TN} - FN\times FP)}{(\text{TP}+\text{FP})\,(FP+TN)+(TP+FN)(FN+TN)} \quad (9)$$

where TP, FP, TN, and FN are the numbers of true positive, false positive, true negative, and false negative pixels, respectively. C and N are the total number of classes and observations.

H. Comparison models

B-Mamba is evaluated against a range of state-of-the-art change detection models. We focus on change detection approaches rather than semantic change detection models, as the latter typically predict both semantic labels and change maps, whereas B-Mamba directly outputs the final change map. The comparison includes CNN-based Siamese Attention U-Net (Bi-AU-Net) [40], SNUNet-CD [41], and the hybrid Transformer-CNN based TransUNet-CD [42], and ConvFormer-CD [43]. Transformer-based baselines include BIT [44], ChangeFormerV6 [45], the fully Transformer Network (FTN) [46] and BiFA [47]. We also compare against the Mamba-based ChangeMamba-B model [31], CDMamaba [48] and HAM-CD [49]. For completeness, we include the Attention U-Net [28] with bi-temporal inputs for direct comparison, as well as its best-performing configuration using spectral bands and indices with OHEM loss.

I. Experimental settings

All models were implemented in PyTorch and trained on an NVIDIA GeForce RTX 3090 GPU with 24 GB memory. Data augmentation included random rotations and flips. Input images were normalized to a [0, 1] range on a per-channel basis. Models were trained for 200 epochs with a batch size of 6 using the AdamW optimizer and a cosine annealing learning rate schedule. The initial learning rate was set to 1×10^{-3}, with a weight decay of 5×10^{-4}. The KaiMing (He) initialization scheme was applied. Model checkpoints were selected based on the best validation mIoU across all burn severity classes.

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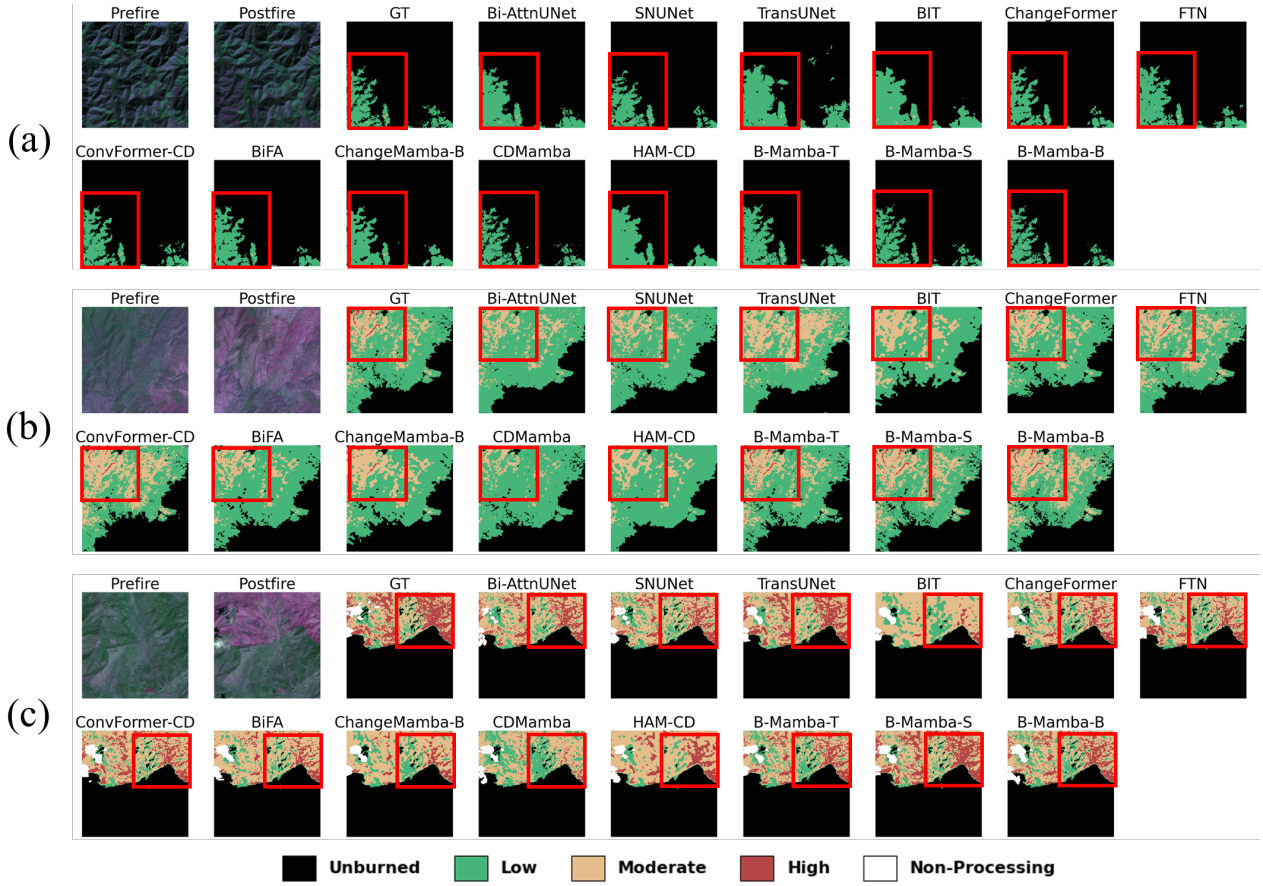


Fig 6. Qualitative comparison of burned area severity mapping on Landsat-BSA test samples. Each row shows pre-fire and post-fire images, ground truth (GT), and predictions from baseline and BS-Mamba models. Red boxes indicate challenging regions. BS-Mamba models demonstrate improved accuracy and spatial detail. Pre and post fire image is the composite of SWIR2, NIR and SWIR1 bands.

III. Results and Discussion

A. Quantitative performance of B-Mamba

Table III summarizes the quantitative comparison between B-Mamba variants and state-of-the-art baselines. Across all metrics, B-Mamba achieves consistently higher performance, indicating improved overall accuracy and class-wise burn severity segmentation. Among the three variants, B-Mamba-S attains the best overall results, with an mIoU of 79.31% and an OA of 95.70%. Relative to the widely used Attention U-Net [28], B-Mamba-S improves mIoU by 3.90% points on the same benchmark, suggesting gains beyond incremental architectural modifications. Moreover, B-Mamba-S also outperforms the best-performing Attention U-Net configuration reported in [28], which uses a combination of spectral bands and indices with OHEM loss and achieves a mIoU of 78.32%, further demonstrating the effectiveness of the proposed model.

Importantly, the advantage of B-Mamba-S is not limited to dominant classes. The model shows more balanced performance across severity categories, achieving IoU values of 78.09% for high severity and 72.33% for moderate severity, while maintaining competitive accuracy for low-severity areas. This behavior is important in burn severity mapping because minority classes often correspond to ecologically and hydrologically consequential conditions but are underrepresented during training. In contrast, several baselines exhibit larger class-wise variability, consistent with a tendency to favor majority classes at the expense of rarer severity levels.

Comparisons with transformer-based and Mamba-based baselines further highlight the robustness of B-Mamba. B-Mamba-S outperforms vision mamba based CDMamba by 3.71% in mIoU and 2.11% in Kappa, while also achieving higher OA and FWIoU. Notably, ChangeMamba-B, despite sharing the VMamba backbone, performs substantially worse, with an mIoU of only 69.00% and a Class 2 IoU of 52.71%, indicating that B-Mamba's gains are not attributable to the backbone alone, but rather to task- specific design choices such as hierarchical change fusion and the decoder reconstruction strategy. B-Mamba-S also exhibits a 4.2% higher mIoU compared to the transformer-CNN based ConvFormer-CD.

Inference-time results show that B-Mamba-S introduces moderate computational overhead (70.6 ms per image) relative to lightweight CNN baselines. Inference-time results reveal an accuracy-efficiency trade-off among the B-Mamba variants. B-Mamba-S achieves the highest mIoU (79.31%) with an inference time of 70.6 ms per image, whereas B-Mamba-T reduces the inference time to 50.8 ms at the cost of only 0.58 percentage points in mIoU. In contrast, B-Mamba-B requires 92.2 ms without improving performance over B-Mamba-S.

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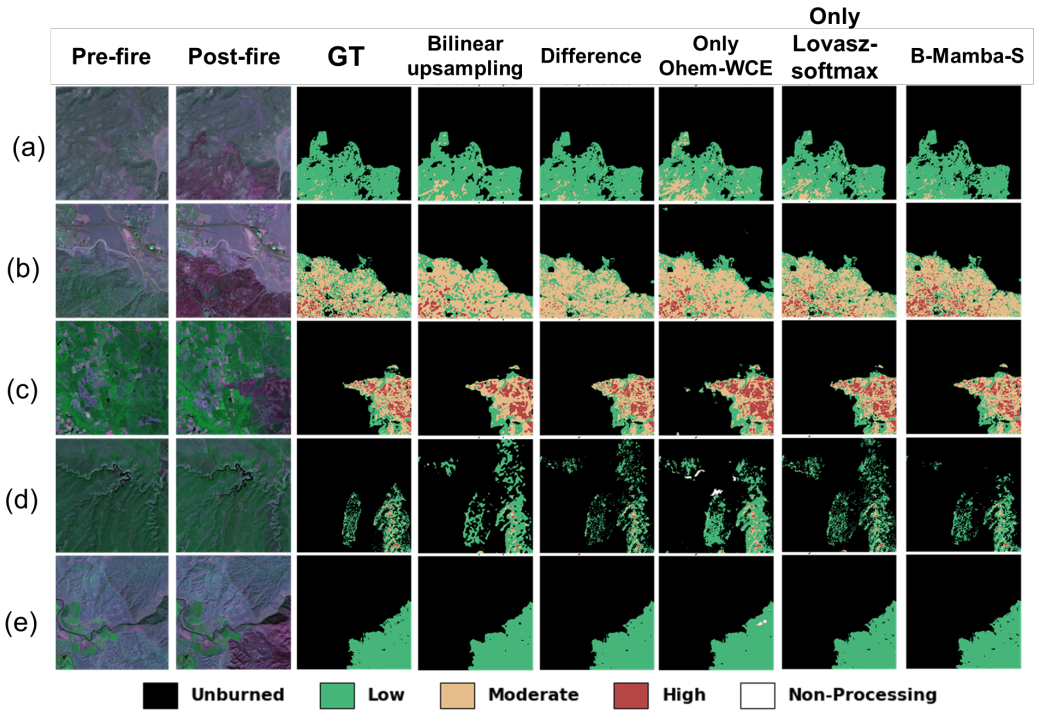


Fig 7. Representative qualitative results from the ablation study comparing bilinear upsampling, difference, OHEM-WCE, Lovász-softmax, and the proposed B-Mamba-S. From (a)-(e) illustrate that the B-Mamba-S model yields more spatially consistent burn severity maps, with reduced omission and commission errors between classes and improved delineation of fragmented burned patches relative to other variants.

For operational deployment, this trade-off should be assessed across the complete workflow rather than inference alone. The reported times represent forward passes on fixed-size patches, while preprocessing, data transfer, and post-processing also affect total latency in Landsat-based burn-severity mapping. B-Mamba-S prioritizes predictive performance, whereas B-Mamba-T provides a lighter alternative for computationally constrained settings.

B. Qualitative comparison

Qualitative results for representative test samples (Fig. 6) corroborate the quantitative findings. Across diverse scenarios including low-severity–dominated scenes, mixed-severity landscapes, and high-severity–dominated events, B-Mamba produces maps that more closely align with reference labels in both spatial extent and severity differentiation.

In particular, B-Mamba yields sharper delineation between moderate- and high-severity classes, which are frequently confused by baselines. CNN-based models such as SNUNet-CD and TransUNet-CD often produce fragmented or overly smoothed predictions in heterogeneous burn scars. ChangeFormerV6 and ChangeMamba tend to generate smoother outputs but frequently blur the boundary between moderate and high severity, leading to systematic confusion. While CDMamba performs comparatively well baselines for low severity class, it still does not consistently capture fine-scale spatial heterogeneity in complex burn patterns like in mixed or high severity dominant region.

These qualitative improvements are closely linked to the patch-expanding reconstruction strategy used in B-Mamba. Compared with bilinear interpolation, patch expansion preserves local spatial relationships during upsampling and reduces excessive smoothing, which supports sharper boundaries and higher segmentation fidelity. This observation is consistent with prior findings that learnable upsampling mechanisms can improve boundary quality in remote sensing segmentation tasks [50]– [52]. Overall, the results suggest that decoder design plays a critical, yet often underemphasized role in translating long-range contextual modeling into spatially accurate severity maps.

C. Ablation study

Ablation experiments (Table IV) quantify the contributions of key components. Replacing patch expansion with bilinear upsampling substantially degrades performance, reducing mIoU to 71.21% (an 8.10 percentage-point decrease relative to B-Mamba-S). The decline is most pronounced for the moderate-severity class, where IoU decreases by 16.90% (from 72.33% to 55.43%). These results indicate that bilinear upsampling suppresses fine-scale spatial variability, which is essential for resolving nuanced severity transitions.

Evaluating fusion strategies shows that simple differencing alone is competitive but inferior (mIoU = 78.74%). Although differencing emphasizes gross spectral change, it lacks the capacity of the proposed cross-scan fusion to model contextual interactions between pre- and post-fire states. Visual results (Fig. 8) further indicate that the full fusion strategy better preserves subtle change boundaries and reduces confusion between burned and unburned regions.

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TABLE IV

ABLATION STUDY ON THE PERFORMANCE OF DIFFERENT COMPONENT AND LOSS FUNCTIONS COMBINATION FOR B-MAMBA-S MODEL. (ALL VALUES ARE IN PERCENTAGE). THE BOLD VALUES SHOW THE BEST SCORE WHILE THE UNDERSCORE SHOWS THE SECOND-BEST RESULT

Strategy	***mIoU***	***OA***	***Kappa***	***IoU per Class [0, 1, 2, 3, 4]***
Bilinear Upsampling	71.21	93.21	83.94	[95.34, 71.19, 55.43, 65.55, 67.79]
Difference	78.74	95.57	89.45	[97.03, 80.43, 71.47, 77.28, 67.49]
Only OHEM Weighted CE	72.73	93.99	86.19	[95.48, 75.79, 68.61, 75.94, 47.83]
Only Lovász-Softmax	78.08	95.48	89.24	[97.07, 80.34, 69.77, 76.06, 67.15]
B-Mamba-S	**79.31**	**95.70**	**89.77**	[**97.10**, **80.94**, **72.33**, **78.09**, **68.06**]

Loss-function ablations underscore the importance of task-specific optimization under class imbalance in comparison with Table I. Training with OHEM-WCE alone reduces mIoU and Kappa, with underrepresented non-processing/cloud class poorly resolved (IoU 47.83%). Lovász–Softmax alone improves boundary quality but remains below the composite objective. Hu et al. [28] evaluated focal, Dice, Lovász–Softmax, cross-entropy, and OHEM losses on Landsat-BSA, with OHEM achieving the highest mIoU. In B-Mamba-S, however, OHEM-WCE alone did not achieve the best performance.

The combined OHEM-WCE + Lovász–Softmax loss yields the most consistent gains across classes (moderate severity 72.33%, high severity 78.09%, and non-processing/cloud 68.06%). This confirms that the two loss terms are complementary in handling the class imbalance, with OHEM-WCE emphasizing hard, under-represented pixels and Lovász–Softmax stabilizing the boundary IoU.

D. Large-scale wildfire mapping and generalization

To assess generalization beyond Landsat-BSA, we evaluate B-Mamba-S on five additional wildfire events spanning diverse geographic and climatic contexts, including non-CONUS regions (Table V). To convert sensor-specific TOA imagery into RGB consistent with our training data, we applied a per-channel 100% linear min-max stretch, rescaling each band's full dynamic range to 8-bit (0–255). For the Alaska and Washington cases, the analysis utilized the Sentinel-2 TOA data provided within the MTBS product, which was already resampled and co-registered to 30 m to match the MTBS reference grid. These two events provide a direct evaluation of B-Mamba-S on Sentinel-2 imagery, separate from the Landsat-based benchmark. The consistent performance gains in both cases (Table V) show that the model is robust across different sensors. Across these events, B-Mamba-S achieves higher average performance than CDMamba, with a 9.35% improvement in mIoU and consistent gains in OA and FWIoU. B-Mamba-S outperforms CDMamba in all the five events. The largest margin is observed for the Washington Vantage Highway fire, where B-Mamba-S improves mIoU by 15.99%, with consistent gains across nearly all severity classes. Improvements are also evident for the Alaska White Mountain Creek fire, the Hawaii Lahaina fire, and the Texas GATE 5 fire.

TABLE V

QUANTITATIVE COMPARISON BETWEEN THE PROPOSED B-MAMBA-S MODEL AND THE OTHER SOTA MODEL FTN FOR BURN SEVERITY MAPPING ACROSS FIVE WILDFIRE EVENTS. THE HIGHEST SCORE FOR TOTAL EVENTS IS HIGHLIGHTED IN BOLD

Wildfire	Model	mIoU (%)	mIoU (w/o) (%)	OA (%)	FWIoU (%)	IoU per Class (%)
Alaska: White Mountain Creek (2018)	B-Mamba-S	64.60	80.75	96.56	93.83	[99.38, 74.02, 71.66, 77.93, 0]
	CDMamba	60.90	76.12	95.37	92.12	[99.51, 74.36, 59.83, 70.77, 0]
Hawaii: LAHAINA (2023)	B-Mamba-S	42.93	85.86	98.25	97.75	[99.31, 72.40, 0, N/A, 0]
	CDMamba	30.48	76.19	97.30	95.48	[98.00, 54.38, 0, 0, 0]
Oregon: BOOTLEG (2021)	B-Mamba-S	64.96	81.19	94.40	90.12	[96.03, 68.15, 76.04, 84.55, 0]
	CDMamba	56.26	70.32	91.40	85.97	[96.55, 68.11, 58.14, 58.49, 0]
Texas: GATE 5 (2020)	B-Mamba-S	58.39	-	98.71	97.80	[99.23, 88.68, 79.49, 0, 24.55]
	CDMamba	55.04	-	98.94	98.19	[99.32, 91.28, 83.54, 0, 1.06]
Washington: Vantage Highway (2022)	B-Mamba-S	60.28	75.35	97.19	94.72	[98.06, 86.68, 56.23, 60.43, 0]
	CDMamba	44.29	55.37	94.94	90.71	[96.93, 76.94, 10.83, 36.76, 0]
Average	**B-Mamba-S**	**58.74**	—	**97.02**	**94.85**	[**98.40, 77.99, 56.69, 55.73, 4.91**]
	CDMamba	49.39	—	95.59	92.49	[98.06, 73.02, 42.47, 33.21, 0.21]

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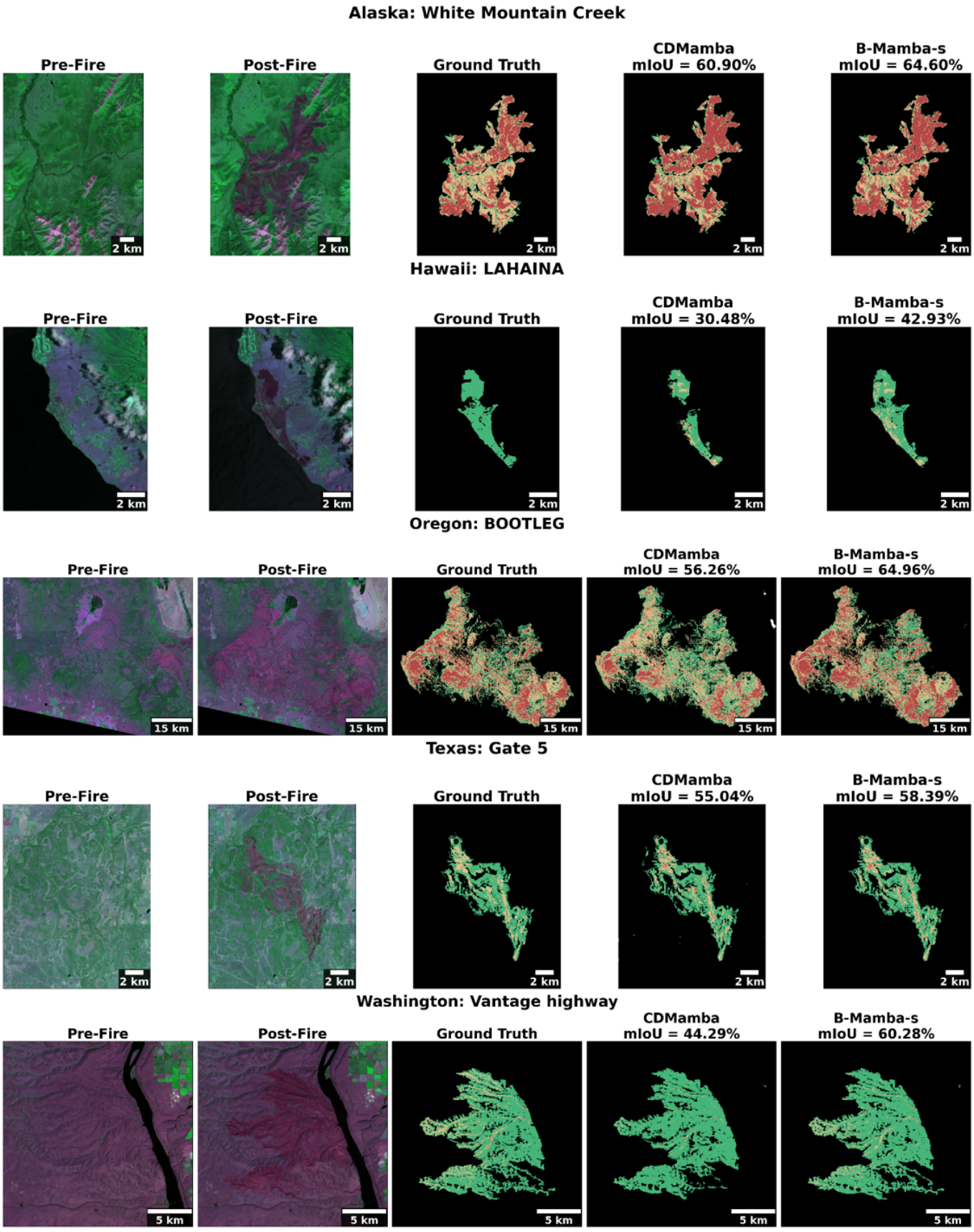


Fig 8. Qualitative comparison of burn severity mapping results for five wildfire events: Alaska: White Mountain Creek (2018), Hawaii: Lahaina (2023), Oregon: Bootleg (2021), Texas: GATE 5 (2020), and Washington: Vantage Highway (2022).

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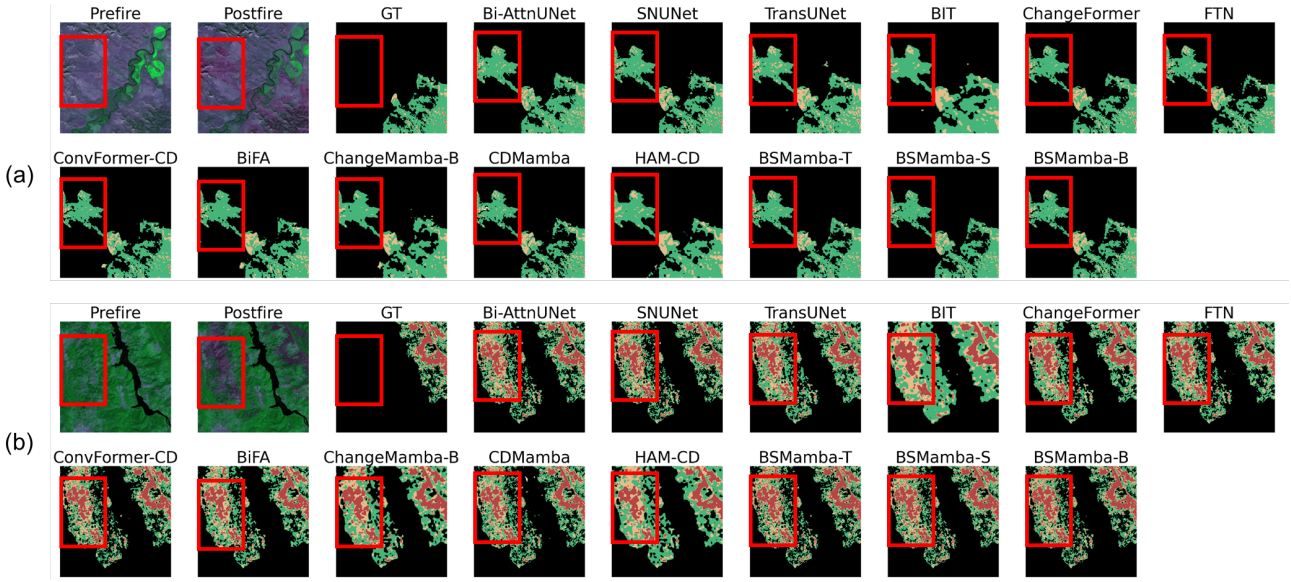


Fig 9. Error presented in the ground truth (GT) dataset of the Landsat-BSA dataset, which affected the model performance in (a) and (b).

Although limited to five events, these cases were selected to be maximize diversity across acquisition conditions. Each occurs in a different year and location, including cases outside the CONUS domain, and they span different vegetation types, fire regimes, and sensor sources. The consistent gains across these conditions suggest a degree of out-of-domain robustness in geography, timeline, and sensor source.

Across the five events, Class 4 (cloud/non-processing) IoU is often near zero, largely because this class is rare or absent in the ground truth for several scenes (e.g., Hawaii), not because of systematic model failure. To isolate this effect, we report mIoU(w/o), excluding only classes absent or negligible in the ground truth, while still counting classes a model missed despite their presence. Under this corrected metric, B-Mamba-S consistently outperforms CDMamba, with the largest gains where multiple classes are absent, such as Hawaii and Washington, confirming the advantage reflects real performance on present classes, not scoring artifacts.

E. Limitations and future directions

A key limitation of the reference labels used for this study arises from how MTBS handles phenological or land cover inconsistency between pre- and post-fire imagery. MTBS severity classes are derived from manually selected dNBR/RdNBR thresholds, allowing an expert analyst to recognize when apparent spectral change reflects seasonal vegetation state rather than fire effect, and to adjust the threshold or image-pair selection accordingly [16], [53]. Because phenological and other non-fire land cover changes can produce spectral change patterns similar to actual burn signal, these errors persist across all evaluated models, including B-Mamba, which along with all benchmarks failed to correctly handle such regular, non-fire change in several scenes.This pattern is visible in Fig. 9, where misclassifications cluster in areas later confirmed to involve seasonal vegetation differences between image pairs. Incorporating phenological information into the modeling framework, as in [54], may help future models distinguish these cases more reliably.

Additionally, uncertainty inherent in the MTBS product itself constrains the achievable performance of any model trained on it as an operational substitute. Independent assessments cited within Picotte et al. [55] report MTBS burned area commission errors ranging from 4.3% to 15.5% in the northern Great Basin and as high as 46.4% in the Pacific Northwest, arising in part because unburned islands within the fire perimeter are not removed from the mapped boundary.

Picotte et al. [55] further show that, although revised MTBS mappings did not differ significantly from original products in overall burned area, the number of pixels assigned to each severity class did differ significantly, with spatial overlap between original and revised classifications ranging from 54% for low severity to 82% for high severity pixels.

This indicates that severity class assignment, rather than burned area delineation, is the primary source of label instability in the MTBS record, and is also the component of the reference data most directly used for supervision in this study. Because MTBS nonetheless remains the most spatially and temporally extensive operational burn severity record available, and because the present objective is to streamline its production through an automated, scalable framework, it was retained here as the reference label set. Any deep learning model trained on MTBS labels therefore inherits an upper bound on accuracy set by the uncertainty of the reference itself, independent of model architecture.

Calibrating against CBI, the field-based standard underlying MTBS thresholds, could reduce reference-label uncertainty, but its sparse, uneven plot-scale coverage limits direct use in this study's multi-region, pixel-wise setting. Future work on CBI-based calibration (e.g., regression-converting dNBR/RdNBR to CBI-equivalent severity) may be used to improve the ground

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truth as the performance of the deep learning models are also highly dependent of the uncertainty of the ground truth label.

IV. Conclusion

This study shows that Mamba-based state-space architectures can provide an effective and scalable approach for large-scale burn severity mapping using medium-resolution satellite imagery. The proposed B-Mamba model consistently outperforms state-of-the-art convolutional, transformer-based, and existing Mamba-based baselines across multiple metrics, improving mIoU, overall accuracy, and class-wise segmentation performance. Notably, the gains extend beyond dominant classes to the moderate- and high-severity categories, which are often underrepresented yet critical for post-fire impact assessment.

Beyond accuracy, the results highlight how architectural choices translate long-range contextual modeling into spatially coherent severity maps. In particular, bilinear upsampling is inadequate within the proposed framework because it suppresses fine-scale spatial variability that is essential for delineating burn severity. Replacing bilinear interpolation with a learnable patch-expanding reconstruction strategy substantially improves boundary definition and class separation, underscoring the importance of decoder design in remote sensing segmentation.

Despite Several challenges remain. Apparent false positives persist in areas affected by non-fire disturbances (e.g., prior burns, forest degradation, and seasonal land-cover changes), reflecting limitations in reference labeling and the spectral ambiguity inherent to medium-resolution imagery. These issues motivate data-centric improvements, including more consistent ground truth and better representation of non-fire change processes in training datasets.

Overall, the findings suggest that Mamba-based architectures are a promising direction for operational burn severity mapping across diverse regions. Future work should evaluate integration with higher-resolution sensors (e.g., PlanetScope) and investigate region-adaptive training strategies to improve transferability. In addition, combining Mamba-based models with foundation-model pretraining and interpretability methods may facilitate broader adoption in operational wildfire monitoring and decision-support systems.

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This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3746053

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This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3746053



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